
Land, Labour and Livelihoods: Unearthing the Perspectives of the Gender Pay Gap in Agri-Food System

Dr. Kanika Pandey, Dr. Sakshi Bhatt, Dr. Soumyadeep Thakur, Alionya Lammingtone Dodo

Introduction

Gender is a major factor in determining economic rewards, decision-making power, labor division, and resource access in agrarian communities. The term “Sex” and “Gender” are frequently used synonymously, they are conceptually different notions that serve as the basis for comprehending gender-based disparities in economic systems, including agriculture. The biological and physiological traits that set males and females apart are referred to as *sex*. The roles, attitudes, duties and power dynamics that societies assign to men and women referred to as *gender*. In developing nations, one of the main sources of employment, especially in rural areas, is the agri-food system. Both men and women participate in food systems, but the nature and extent of their participation varies depending on the structure of the economy and gender norms. Women play important roles at different nodes of both agricultural and off-farm value chains, but in many countries their contributions are either underestimated or limited by prevailing societal norms or gender-specific barriers (Quisumbing *et al.* 2021). Women are actively involved in a range of roles from production and processing to retailing and consumption; they grow and manage crops, tend livestock, work in agribusinesses and food retailing, prepare food for their families, and much more (Malapit *et al.*, 2020). However, women's contributions to food systems are frequently overlooked or underappreciated and when women are not acknowledged as "real farmers," their environmental and farming endeavors are either hidden or seen as an extension of domestic duties. Women still face persistent economic disadvantages, most notably the gender pay gap, which is defined as the systematic difference in earnings between men and women performing comparable work or contributing similar labor inputs. According to estimates from the International Labour Organization's *Global Wage Report 2018/19*, women earn approximately 20 percent less than men across the world, with larger median gender wage gaps observed in lower-income countries (ILO, 2018). In the agri-food system, this gap is frequently wider than in non-agricultural sectors due to structural inequalities, informality of work, and deeply ingrained socio-cultural norms. In order to

achieve gender equality among lower-income rural households, it is imperative to address gender wage discrepancies in agricultural work.

The gender pay gap in agriculture is a reflection of larger differences in access to productive resources, education, skills, technology, and decision-making authority rather than just unequal salaries. Because of these differences, women's productivity and negotiating power are restricted, which perpetuates cycles of poverty and food insecurity. Therefore, closing the gender pay gap is essential for attaining inclusive agricultural growth, enhancing rural livelihoods and fulfilling international commitments like the Sustainable Development Goals (SDGs), especially SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 5 (Gender Equality) and SDG 8 (Decent Work and Economic Growth). However, it is important to understand the gap and estimate the share due to gender discrimination and further frame policies aimed at tackling the gender wage gap. In order to develop gender-equitable labor market policies, it is crucial to comprehend the factors that contribute to the gender wage gap and the role that gender-based discrimination plays at various stages of structural development. In this chapter, the factors responsible for the gender wage gaps and policies are discussed that need to be embedded within food systems for women empowerment and gender equality.

2. Historical evolution of Women's Role in Agriculture

The historical evolution of women's roles in agriculture is characterized by a transition from being the primary, albeit often unacknowledged, architects of early cultivation to becoming "silent heroes" within rationalized systems and finally trying to emerge as professional scientists and leaders. According to Inhetveen, 1998 in prehistoric and ancient times, women were the principal contributors to agricultural progress, accumulating thousands of years of botanical knowledge and innovating plant varieties and tools, though these were frequently documented as "no-name products". While Graeco-Roman myths stylized women as powerful "agrarian goddesses" like Ceres, the social reality was governed by a gender construct that viewed women as too weak for the outdoors, often erasing their actual labor from historical records (*Scheidel 1990; Müller and Klemm 1988*)

By the 16th century, European "husbandry literature" began to indirectly acknowledge women's comprehensive knowledge, identifying them as "female bridges" who disseminated agricultural innovations across regions through marriage and migration. Figures like Anna Electress of Saxony emerged as elite pioneers, managing vast estates and conducting sophisticated experiments in alchemy and veterinary medicine. The 20th century trying the return of women to professional agricultural

leadership. Pioneers like Margarethe von Wrangell, who became Germany's first female full professor in 1923, and Lady Eve Balfour, who founded the Soil Association, demonstrated that women were central to both academic science and the nascent ecological farming movement. In the United States, this professionalization was reflected in education as well; from 1929 to 1969, women moved from being portrayed in supportive domestic roles such as "FFA Queens" or participants in home economics to operating heavy machinery during World War II and ultimately gaining full membership in the National FFA Organization in 1969.

In a colonial context, such as Zimbabwe, women's roles underwent a cycle of status and marginalization. While they were the backbone of production in the pre-colonial era, the colonial period transformed them into "invisible subsistence producers" as male migration to mines and cities left women as de facto farm managers without legal rights to credit or extension services (Chinyemba *et al.*, 2006). In Indian Women in Development Perspective, Pande 2007 argues that in the Indian agriculture context, women were marked by a transition from unacknowledged cultivators to an increasingly pauperized labor force. Historically, agricultural policy has been dominated by the false view that "farmers are men" and women are merely "housewives,". Following independence, early Five-Year Plans (1951–1979) adopted a welfare approach, focusing on productivity while viewing women's roles as secondary. The Green Revolution (1965–66) subsequently increased women's manual workloads, such as weeding, while destroying the biodiversity they traditionally managed.

According to the census report 1981 Statistically, a dramatic shift occurred between 1951 and 1991: female "cultivators" fell from 45.40 percent to 34.22 percent, while "agricultural laborers" surged from 31.90 percent to 43.93 percent. Post-1991 globalization exacerbated this by transferring seed control to multi-national corporations, further eroding women's traditional knowledge and control. While women currently perform 61.00 percent of all farm work, they are often rendered "invisible" by statistics that categorize them merely as housewives. It was not until the Tenth Plan (2002–2007) that policy explicitly shifted from welfare toward empowerment. (Bollikonda *et al.*, 2017)

Between 2010 and 2025, the status of women in Indian agriculture underwent a significant shift characterized by a "forced back" migration to cultivation, especially during the 2020–2022 pandemic-related economic slowdown. While the 2011 Census reported that nearly every second agricultural laborer was a woman, by 2022, women represented 62.80 percent of the total agricultural workforce, compared to only 38.10 percent of men. This trend of "*feminization of agriculture*" continues in modern contexts like Nepal, where women now represent 50.90 percent of the agricultural population

due to ongoing male migration for non-farm jobs, leaving women to manage food security while carrying a "triple burden" of production, reproduction and exploitive societal norms. This "feminization of agriculture" is primarily driven by male out-migration for non-farm jobs, leaving women to manage food security while carrying a "triple burden" of production, reproduction and exploitive societal norms. Senapati and Mukhopadhyay (2024) stated that despite their dominance, women face persistent structural inequality. In 2023, the gender pay-gap remained high at 27.00 percent, with women earning significantly less than men for similar labor. Furthermore, during 2018–2022, female self-employed workers experienced a higher negative rate of earnings growth compared to their male counterparts. While initiatives like Self-Help Groups (SHGs) and the National Rural Livelihoods Mission have improved access to credit, women still struggle with limited land ownership and a lack of decision-making power, meaning true empowerment in the sector is still a work in progress. Despite these advancements, the legacy of women as "silent heroes" persists, as they continue to perform the bulk of agricultural labor worldwide while often lacking the same institutional support as their male counterparts.

3. Conceptual Understanding: Gender Pay Gap in the Agri-Food System

The Gender Pay Gap is the systematic difference between the average earnings of men and women, with global estimates from the International Labour Organization indicating that women earn approximately 20.00 percent less than men. According to the Benali 2025, the gender pay gap can be systematically examined using wage decomposition techniques like the Kitagawa–Oaxaca–Blinder (KOB) method (*Blinder, 1973; Kitagawa, 1955; Oaxaca, 1973*), which separates observed wage differentials into two core components. The first is the endowment (explained) Effects which captures the share of the wage gap caused by differences in observable characteristics between men and women, such as education levels, age, work experience, and household composition. The second is the structural (unexplained) effect, which represents the portion of the gap that persists even when men and women possess similar characteristics. This unexplained component reflects unequal returns to comparable endowments such as women receiving lower wages than men with the same educational attainment and is commonly interpreted as evidence of labor-market discrimination or gender-biased institutional structures.

In the specific context of the agri-food system this gap reflects a "wedge" between the marginal product of labor and the actual market wage received, a disparity that the sources indicate is generally larger

for women than for men. Agriculture is very often portrayed as and perceived to be a masculine activity (Elias *et al.*, 2018). Women's labor engagement is significant but underappreciated in the agri-food system, which includes a wide range of activities such as agricultural production, livestock rearing, fisheries, post-harvest management, agro-processing, marketing, and food distribution. It is commonly stated as women earning less per hour or unit of work than males. The disparity between male and female earnings can be expressed either in absolute terms (difference in daily or monthly income) or in relative terms (% difference). It is crucial to evaluate the magnitude of gender wage disparities and comprehend the factors that influence them at various phases of economic development. The gender pay gap is like *two people running the same distance on a track, but one runner is assigned to a lane filled with invisible hurdles*. Even if both runners have the same training and stamina (endowments), the runner in the lane with hurdles will always finish with a slower time and a smaller reward (structural discrimination) because the track itself was not built to be equal.

This variation is particularly evident across the agri-food system, which is broadly divided into primary agriculture, the wider agri-food system encompassing processing and services, and sectors outside the agri-food system, with gender pay gaps varying distinctly across these stages. In primary agriculture, particularly in developing countries such as India, women often work as invisible or casualized laborers, resulting in a gender pay gap of roughly 27.00 percent as of 2023. According to a periodic labour survey report (2024), a similar pattern is observed in Malawi, where rural female wages were found to be 35.00 percent lower than males. However, evidence from the lowest-income countries, including Ethiopia and Uganda, suggests that gender wage gaps in primary agriculture are sometimes statistically insignificant or lower than those in middle-income economies, a pattern that changes as countries undergo structural transformation and labor markets become more differentiated (Palacios *et al.*, 2015). As the employment shifts beyond the farm into a broader Agri-food system covering food processing, manufacturing, and related services the gender wage gap tends to widen further. At this stage, both endowment effects, such as disparities in education and work experience, and structural effects, including discrimination and unequal returns to labor, play nearly equal roles in sustaining wage inequalities. In contrast, sectors outside the agri-food system generally exhibit smaller gender pay gaps, largely driven by differences in human capital rather than overt structural disadvantages. As economies move toward more formal, skill-intensive employment, the mechanisms underlying wage gaps increasingly differ from those observed in the informal and labor-intensive agricultural settings typical of developing countries.

3.1 Difference in developed and developing countries Gender pay Gap

The patterns of Gender pay gap differ markedly between developing and developed countries, reflecting broader structural and historical dynamics. In Developing Countries, the larger median wage gaps are observed in lower-income countries. A primary driver is pauperization; for example, in India, women have transitioned from "cultivators" (landowners) to "agricultural laborers" (wage earners), losing control over production and becoming dependent on low-paying seasonal work. In these regions, unexplained factors (discrimination) often account for over 80–90 percent of the gap. Women's wages also tend to be lower than men's in LMICs, even for the same activities (*de Brauw et al., 2021; Kruijssen et al., 2018*), pointing to potential gender-based discrimination in agriculture and agrifood systems overall. Poorer rural households also tend to have greater participation and reliance on agricultural wage employment than do wealthier and more-urban households (*Davis et al., 2017; Davis et al., 2010; Winters et al., 2010*). In India, the gender pay gap was recorded at 27.00 percent in 2023, meaning women received only 73.00 percent of what men earned for similar work. In Ethiopia, the gap was found to be roughly 11.50 percent when measured by value, with structural effects (unexplained factors) having a more significant impact than endowment effects.

Whereas, In Developed Countries while the gap has decreased over time in high-income countries, it remains open. Significant gender gaps persist even in highly developed agricultural sectors like the United States, where women often face gender bias when assigned to specific roles like horticulture instead of animal science or welding. A few studies have examined specifically the gender wage gap in agriculture in high-income countries. For example, among farm workers in the United States of America, find gender differences in the types of farm tasks performed and crops cultivated, which helps explain the male–female differential in wages and benefits received. Women earn an average USD 0.95 for every USD 1 earned by comparable male farmworkers, conditional on worker characteristics. A gender wage gap exists for both part-time and full-time workers and for parents (*Fisher et al., 2022*).

4. Determinants of the Gender Pay Gap in Agri food system

The gender pay gap in the agri food system is caused by a complex interplay of institutional, structural, cultural and socioeconomic elements that function at various stages of agricultural value chains. Gender-based wage discrepancies are exacerbated by the fact that agricultural labor markets, in

contrast to formal sectors, are primarily informal, seasonal, and poorly regulated. The following section discusses the main factors that contribute to the gender pay disparity in the agri-food system.

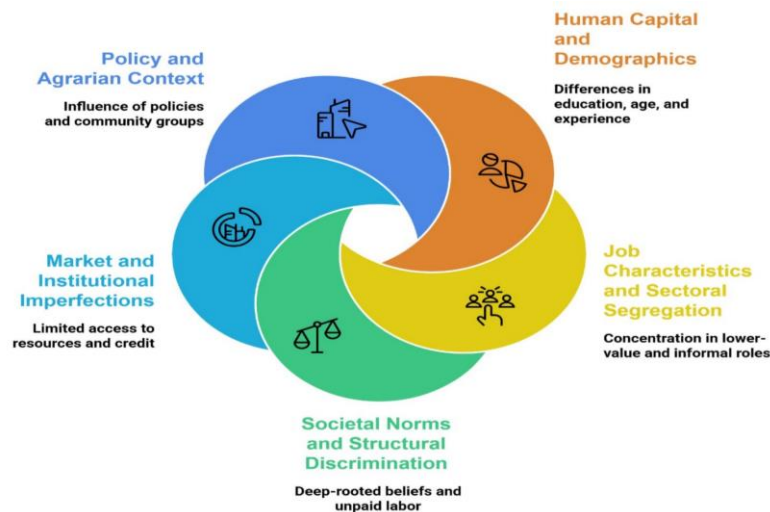


Figure 1- Factors Contributing to Gender Pay Gap in Agri Food System

4.1 Human capital and individual demographics serve as primary "explained" determinants of the wage gap. Gender disparities in education, age, and work experience significantly shape earnings outcomes, as rural women have historically faced lower investments in schooling and skill development than men.(Blau and Kahn, 2017; Weichsel baumer and Winter-Ebmer, 2005). These disadvantages translate into lower productivity, occupational mobility and lifetime earnings.Limited formal education and literacy often confine women to low-skilled, low-paying agricultural roles, while even in off-farm segments where women possess higher education, they continue to receive lower returns than men for comparable qualifications. Life-course dynamics further exacerbate these gaps: although women enter agri-food employment at younger ages, labor force participation frequently declines during childbearing years (25–34), interrupting experience accumulation and reducing long-term earning potential. In addition, household composition and caregiving responsibilities impose severe time constraints, forcing women into part-time or irregular employment. As a result, women work fewer than 40 hours per week on average in agriculture, limiting total income relative to men who are more likely to hold continuous, full-time positions.

4.2 Job characteristics and sectoral segregation are critical drivers of wage inequality within the agri-food system (AFS). Women are systematically concentrated in lower-value node and labor-intensive segments of the value chain, such as on-farm casual labor, petty retail, and small-scale processing, while men dominate more lucrative activities including wholesale trade, transportation,

mechanization, and managerial roles (Fisher *et al.*, 2022; Fremstad and Paul, 2020). Additionally, women are more likely to be engaged in vulnerable and informal employment, often working without written contracts or fringe benefits, which results in a "double burden" of low wages and a lack of social protection which reinforces persistent income disparities within the sector (Fisher *et al.*, 2022; Fremstad and Paul, 2020)

4.3 Societal norms and structural discrimination constitute the "unexplained" portion of the pay gap, where women receive lower returns than men even when they possess identical skills and resources. Deeply entrenched beliefs that farming is primarily a male occupation relegated women to the status of secondary or auxiliary workers, leading to the systematic undervaluation of their labor in markets, institutions, and official statistics. Women also shoulder a disproportionate burden of unpaid domestic and care work, including child care, cooking, and water collection, which creates chronic "time poverty" and constrains their ability to engage in regular, well-paid employment. Restrictive social norms further limit women's mobility, market participation, and access to extension services and decision-making bodies, even when they perform traditionally male agricultural tasks due to male out-migration, thereby perpetuating cycles of marginalization and low remuneration. In some contexts, the "male breadwinner" myth leads to the systematic undervaluation of female labor as "secondary" or "unskilled". (Fao Report, 2023)

4.4 Market and institutional imperfections further intensify the gap by limiting women's access to productive resources and economic opportunities. The "gender asset gap" characterized by women's lack of formal land titles and ownership of livestock prevents them from providing the collateral necessary to access agricultural credit. This restriction limits their ability to invest in modern inputs such as high-yield seeds, fertilizers, and mechanized equipment, which in turn lowers their individual labor productivity and earning potential compared to men who have a monopoly over these technologies. (Chinyemba *et al.* 2006). Labor markets themselves are not gender-neutral often accounts for over 60.00 percent to 70.00 percent of the unexplained wage gap in some regions; discrimination is estimated to account for a substantial share of the unexplained wage gap in several regions, creating a wedge between women's actual wages and the value of their marginal productivity. In some contexts, institutionalized practices, including historically sanctioned wage-setting norms or

decisions by local governance bodies, have explicitly prescribed lower wages for women than for men for equivalent work (Barret *et al.* 2008, Palacios *et al.*, 2015)

4.5 Policy and Agrarian Context play a significant role in shaping the magnitude of gender pay gaps in agriculture. Evidence suggests that higher cropping intensity, often driven by improved irrigation infrastructure, increases labor demand and can enhance women's bargaining power, thereby contributing to narrower wage differentials. Participation in self-help groups (SHGs) has emerged as an important pathway for improving women's access to credit, alternative income sources, and collective bargaining strength, reducing reliance on low-paid agricultural wage labor. Although public employment programs such as the Mahatma Gandhi National Rural Employment Guarantee Program mandate equal wages by law, their effectiveness in reducing gender wage gaps in private agricultural labor markets remains uneven across states. (Kundu *et al.*, 2019).

5. Global Scenario of Gender Pay Gap Analysis

Despite the variation in geographical region, the gender pay gap in agriculture remains intact in all countries. Due to social conventions, informality and restricted land rights, women agricultural workers in developing South Asian and Sub-Saharan African nations make substantially less than men. While women, primarily employed in low-paying, seasonal jobs, wage gaps in Latin America have not been addressed despite greater female participation in agri-food value chains. Gender wage disparities still exist even in industrialized nations like North America and Europe, especially for migrant and temporary female employees. These cross-national differences in agricultural incomes are still reinforced by unequal access to resources, skills, and technology. Below the worldwide perspectives of gender pay gap is presented in tabular form.

5.1 Perspective from Africa

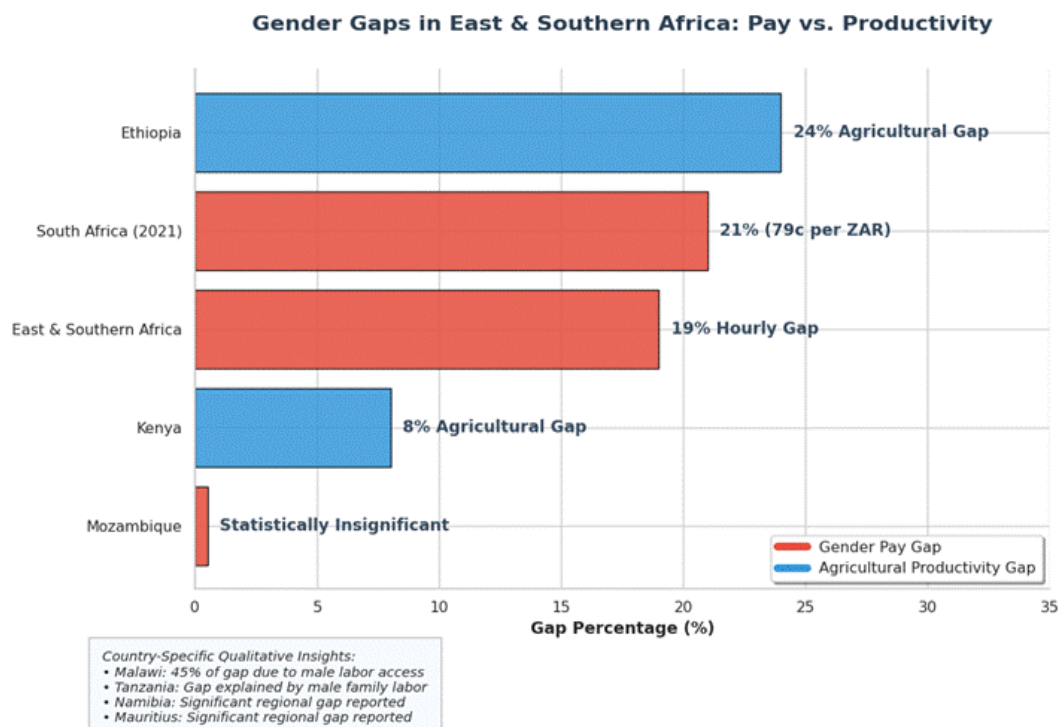


Figure-2 Gender pay gap scenario in Africa

As shown in Figure-2 the difference across wages shows that the *productivity gap*, which reflects disparities in efficiency, technology, and resource utilization, is the difference in output generated per worker or per unit of input between sectors, regions, or groups. On the other hand, the *wage gap*, which is impacted by labor market institutions, abilities, and social issues like discrimination, indicates disparities in workers' earnings even when their productivity may be comparable. Africa reflects the story of women's earnings compared to men's is one of persistent inequality, though the reasons shift from place to place. In East and Southern Africa, for example, women often bring home less money than men, earning about 81 cents for every dollar, largely because they spend fewer hours in paid work (UN Women, 2023). On farms, the gap feels even sharper. In Ethiopia, women's productivity is 24% lower, mainly because they cannot rely on male family labour, and experts say that closing this gap could lift more than a million people out of poverty (UNDP, 2018). Kenya offers a brighter picture, with one of the smallest gaps at just 8%, but in Tanzania and Malawi, the lack of male labour support explains much of the inequality, nearly half of Malawi's gap is tied to this single factor (UN Women Policy Brief No. 11, 2019). Rwanda too struggles with differences in agricultural productivity (UN Women, 2015), while Mozambique shows a different twist: once worker characteristics are taken into account, the pay gap almost disappears (UN Women, 2023). Other countries, like Namibia and

Mauritius, are also highlighted in regional studies for their significant disparities (UN Women, 2023). South Africa’s story is especially troubling, as the gap has grown worse over time, women earned 89 cents for every dollar in 2008, but by 2021 that figure had dropped to 79 cents, driven mainly by labour market segregation (SA-TIED, 2021). Taken together, these patterns reveal how structural barriers like access to labour and opportunities—combine with wage inequalities to keep women at a disadvantage across the continent.

5.2 Perspective from South East Asian Countries

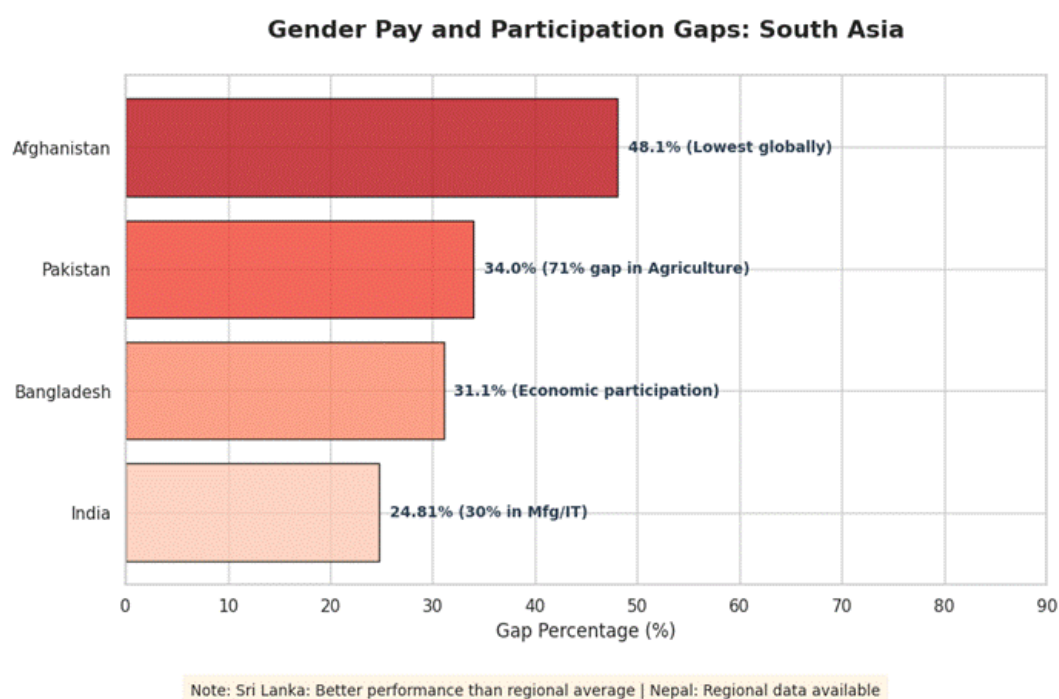


Figure-3 Gender pay gap scenario in East Asia

In South Asia, the gender pay and participation gap remains a major challenge, with wide disparities across countries. Afghanistan has the world’s lowest gender equality ranking, with a 48.1% overall gender gap driven by severe restrictions on women’s participation in education, work, and public life (World Economic Forum, 2023). Pakistan shows one of the steepest gaps, with women earning 34% less overall and men making 71% more in agriculture even after controlling for factors, as women are concentrated in low-wage agricultural labour, livestock, and dairy (ILO Global Wage Report, 2025). India’s wage gap was 24.81% in 2013, rising to 30% in manufacturing and IT, with agriculture, employing two-thirds of the workforce and accounting for most of the inequality with 60% of the pay

gap attributed directly to discrimination (World Bank, 2023). Bangladesh faces a 31.1% economic participation gap, with 71% of its female labour force concentrated in agriculture, reflecting limited diversification of women’s employment opportunities (World Economic Forum, 2024). Nepal highlights similar structural challenges (ORF, 2021). Sri Lanka performs better than the regional average, showing relatively lower gaps (World Economic Forum, 2023). Afghanistan, however, represents the most severe case globally, with a 48.1% overall gender gap, driven by extreme restrictions on women’s participation in education, work, and public life (World Economic Forum, 2023). Overall, the region illustrates both entrenched structural inequalities in agriculture and labour markets, and stark contrasts between countries with modest progress and those facing systemic exclusion of women.

5.3 Perspectives from Latin America

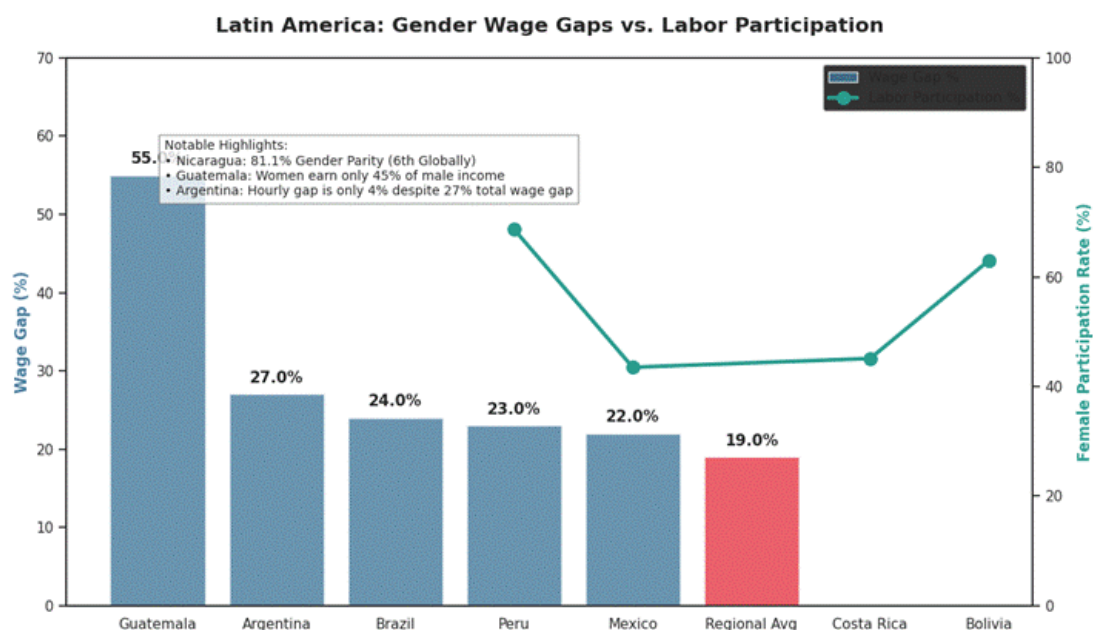


Figure-4 Gender pay gap scenario in Latin America

In Latin America, the gender pay and participation gap shows both progress and persistent inequality. On average, the region has a 19% wage gap and a 17% hourly earnings gap, which is better than the global average of 24% (ILO Global Wage Report 2014–15). Yet, country-level data reveal sharp contrasts: Argentina records a 27% wage gap overall, but among agricultural workers the gap rises to 46%, the lowest ratio in the region (IDB Technical Note, 2021). Brazil follows with a 24% wage gap

across sectors, including agriculture (ILO, 2015; IDB, 2023). Peru shows a 23% gap but also the highest female labour participation at 68.7%, with women concentrated in mixed crops and livestock (ILO, 2015; UN News, 2019). Mexico has a 22% wage gap and lower female participation at 43.5% (ILO, 2015; UN News, 2019). Guatemala stands out negatively, with women earning only 45% of male income—the worst in Latin America (World Economic Forum, 2024). Costa Rica also struggles with low female participation at 45.1% (UN News, 2019). On the positive side, Nicaragua achieves 81.1% overall gender parity, ranking sixth globally despite wage gaps (World Economic Forum, 2023), while Bolivia shows high female workforce participation at 63% (UN News, 2019). Overall, the region demonstrates a mix of high participation in some countries but persistent wage inequalities, with structural barriers in agriculture and labour markets continuing to disadvantage women.

5.4 Perspectives from North America

In North America, the gender pay gap among farmworkers reflects a subtle but persistent inequality. In the United States, even after controlling for factors such as education and experience, women earn about 5–6% less than men, and they are also less likely to receive bonuses or employer-paid health insurance, pointing to structural disadvantages beyond wages alone (Li & Reimer, 2021). Canada shows similar patterns, with studies indicating that women face comparable disparities in pay and benefits, aligning with broader trends observed in developed countries (Gender Equality Agency, 2023). Overall, while the wage gap is narrower than in many other regions, the persistence of unequal access to benefits and incentives highlights that gendered inequities in agricultural labour markets remain an ongoing issue.

5.5 Perspectives from Europe

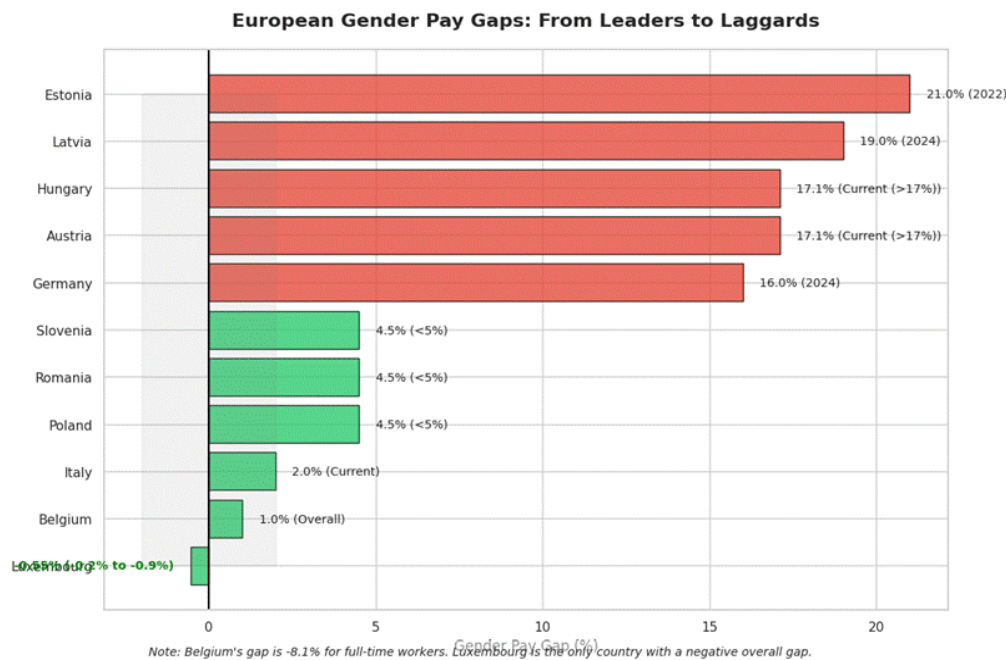


Figure-5 Gender pay gap scenario in Europe

In Europe, the gender pay gap paints a mixed picture, with some countries showing persistent inequality while others have nearly closed or even reversed the gap. Germany remains among the highest in the EU, with women earning 18% less than men in 2023 and 16% less in 2024 (Eurostat, German Federal Statistical Office, 2024). Austria and Hungary also report gaps above 17%, while Estonia consistently records the highest in the EU at around 21% (Eurostat, 2022–2023). Latvia follows closely with a 19% gap (Eurostat, 2024). On the other hand, several countries demonstrate much smaller disparities: Belgium shows only a 1% overall gap, and in some cases women working full-time earn more than men (Eurostat, 2023). Italy’s gap is just 2%, while Poland, Romania, and Slovenia each report gaps below 5% (Eurostat, 2023). Luxembourg even shows a slight reverse gap, with women earning marginally more than men on average (Eurostat, 2022–2023). Overall, while Central and Eastern European countries like Estonia and Latvia struggle with some of the widest gaps, Western and Southern European nations such as Belgium, Italy, and Luxembourg illustrate that

narrowing or eliminating the gap is possible, highlighting both progress and persistent challenges across the continent.

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6. Gender Pay gap in Indian Agriculture

India's agricultural sector demonstrates severe gender wage inequality, with women earning roughly half of what men earn and 60% of this pay gap stemming from pure discrimination rather than legitimate factors like skills or experience, while the overall gender wage gap across all sectors remains between 19-27 percent (ILO India Wage Report 2023; Sattva/PLFS 2019-20; World Bank 2023). The situation has deteriorated significantly in recent years, as unpaid female agricultural workers more than doubled from 23.6 million to 59.1 million between 2017-18 and 2024-25, women's wages actually declined while men's wages grew, and despite increasing female participation in agriculture reaching 44.70 percent, women remain concentrated in lower-paying tasks where wage gaps can exceed 40% compared to male-dominated skilled operations (PLFS 2023-24; UNDP 2025; Mohanty *et al.* 2014). Below in multi-dimensional table the perspectives of gender pay gap in India agriculture has been presented-

6.1 Gender Pay gap in Indian Perspective

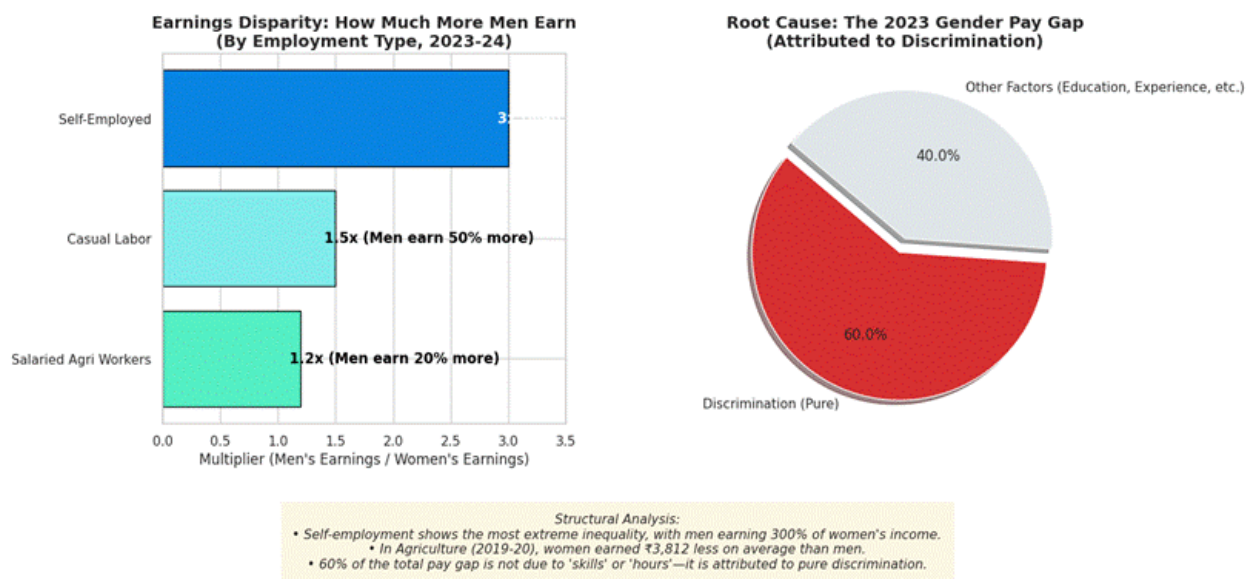


Figure-6 Gender pay gap in Indian perspective

Gender inequality in wages remains a serious issue in India and can be seen across different types of work. On average, women earn about 19–27% less than men across all sectors (ILO India Wage Report 2023), showing that the gap exists widely in the economy. This difference is even more visible in agriculture, where women earn only about half of what men earn, with an average shortfall of ₹3,812 (2019–20, Sattva / PLFS 2019–20). Among self-employed workers, the gap is much larger, as men earn nearly three times more than women (2023–24, PLFS 2023–24). Even in regular and casual jobs, inequality continues, with men earning 1.2 times more in salaried agricultural work (2023–24) and 1.5 times more in casual labour (2023–24). A very important point is that about 60% of this wage gap is not due to differences in skills or work, but due to discrimination itself (2023, World Bank 2023). This clearly shows that gender-based inequality in earnings is not just an economic issue but also a social one that needs attention.

6.2 Gender Pay Gap Perspective on Labour Force Participation

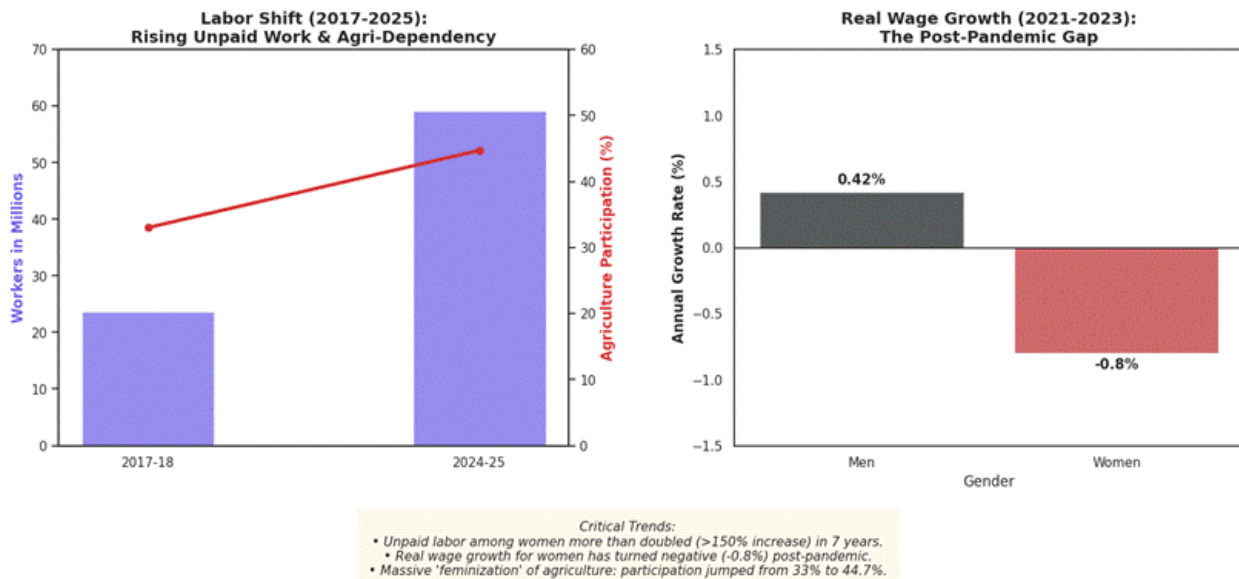


Figure-7 Gender Pay Gap Perspective on Labour Force Participation

Gender differences in labour force participation in India reveal deep structural inequalities in access to work and the type of employment available to men and women. In 2023–24, the Labour Force Participation Rate (LFPR) for women was 41.70%, compared to 78.80% for men, resulting in a large gap of 37.1 percentage points (PLFS 2023–24). A significant share of working women about 44.7%

are engaged in the agriculture sector, while the proportion of men in agriculture is comparatively lower, indicating that women are more concentrated in this sector (PLFS 2023–24, UNDP 2025). The gender gap in participation also varies spatially, being relatively lower in rural areas but higher and widening over time in urban areas (UNDP 2025). Another important concern is the high proportion of women working as unpaid family workers in agriculture, accounting for nearly 50%, whereas this share is much lower for men, highlighting a major form of hidden and unrecognized labour inequality (UNDP 2025). Together, these patterns show that women’s participation in the workforce is not only lower but also more likely to be informal, unpaid, and concentrated in less secure sectors.

6.3 Gender Pay Gap Perspective on Gender-Segregated Agricultural Tasks

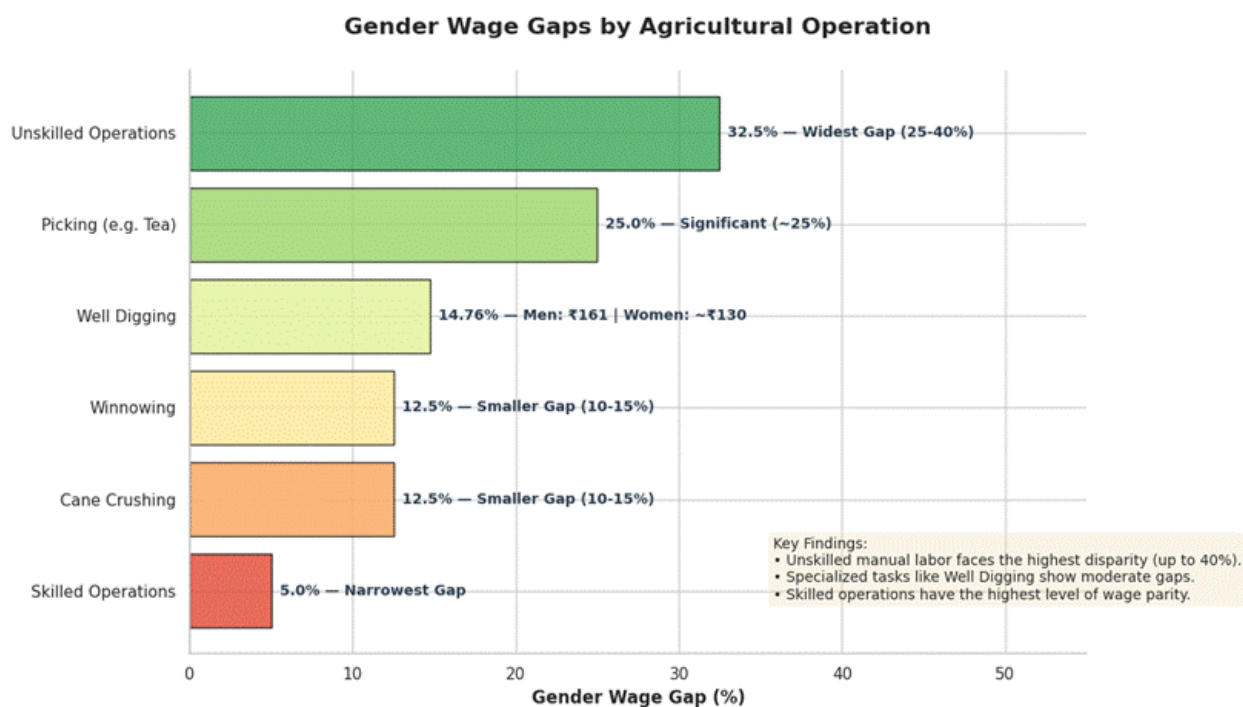


Figure-8 Gender Pay Gap Perspective on Gender-Segregated Agricultural Tasks

Wage differences between men and women in agriculture also vary across specific farm operations, reflecting the nature of work and skill involved. For instance, in well digging, men earn about ₹161 per day while women earn around ₹129–132, resulting in a significant gender gap of about 14.76% (Mohanty et al. 2014). In activities such as picking, including tea leaf plucking, men generally receive higher wages than women, with gaps estimated at around 20–30%. In operations like winnowing and cane crushing, the wage differences are relatively smaller, typically in the range of 10–15% (ILO India

Wage Report 2023). Interestingly, in more skilled agricultural operations, the wage gap tends to narrow, as women's earnings come closer to those of men. However, in unskilled tasks, the disparity becomes much wider, with women earning significantly less and the gap ranging from 25–40%. These patterns indicate that gender wage inequality in agriculture is not uniform but depends on the type of work, with larger gaps persisting in low-skilled and labour-intensive activities.

6.4 State-Wise Agricultural Wage Perspective in India

India's gender pay gap is a complex and deeply rooted issue shaped by social norms, occupational segregation, and structural discrimination. Women are often concentrated in lower-paying sectors such as agriculture and informal work, where their contributions are undervalued and frequently unpaid. Even when they participate in wage employment, they face persistent barriers including unequal access to opportunities, limited recognition of their skills, and workplace cultures that reinforce traditional gender roles. Over time, this has led to a feminization of agriculture, with more women entering the sector but often without fair compensation or security. Regional differences further highlight how cultural attitudes and local economies influence women's earnings, yet across the country the underlying challenge remains the same: women's work is consistently seen as secondary, leaving them with fewer opportunities, lower wages, and heavier burdens of unpaid labour. The persistence of these patterns shows that India's gender pay gap is not just about numbers, rather it reflects systemic inequities that demand structural change in both policy and practice. Below the state-wise scenario of agricultural gender pay gap has been presented-

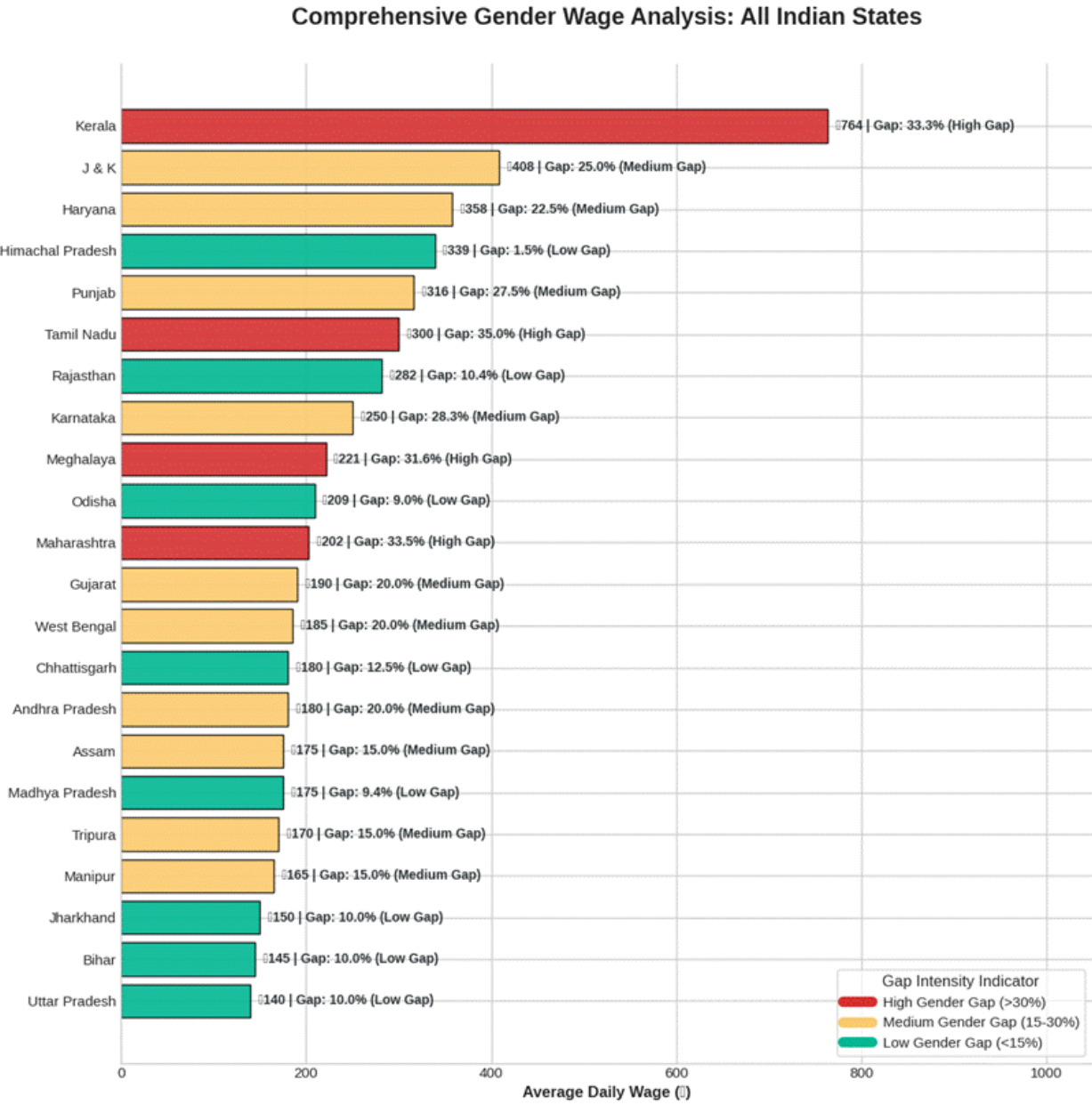


Figure-9 State-Wise Agricultural Wage Perspective in India

State-level patterns of agricultural wages in India show wide regional variation along with persistent gender disparities. Among high wage states, Kerala stands out with very high daily wages for men ₹641.07 in 2015–16 rising to ₹764.3 in 2022–23 while women earned ₹427.50, resulting in women receiving about 66.70% of men’s wages and a gap of 33.30% (Jose 2020, Wire 2024). States like Haryana and Punjab also rank high in wages, though women’s data are limited, with estimated wage ratios of 70–80%, indicating gaps of around 20–30% (Jose 2020, NABARD 2016). Himachal Pradesh

shows near gender parity, with women earning about 98.50% of men's wages and only a 1.50% gap (Jose 2020, NABARD 2016), while Karnataka reflects a wider disparity, with women earning 71.70% of men's wages (NABARD 2016). In moderate wage states such as Rajasthan, the gap is smaller at about 10.40%, whereas Maharashtra and Meghalaya show higher disparities of around 30–37% and 31.60% respectively, despite their differing economic contexts (NABARD 2016). Industrial states like Gujarat also present relatively low wages and variable gender gaps (NABARD 2016, Jose 2020), while states such as West Bengal, Andhra Pradesh, Assam, and Tripura fall into the moderate category with continuing but less clearly defined disparities (NABARD 2016). In contrast, low wage states including Chhattisgarh, Madhya Pradesh, and Odisha exhibit relatively smaller gaps of around 9–15%, with women earning about 85–91% of men's wages (Jose 2020, NABARD 2016), whereas Bihar, Jharkhand, and Uttar Pradesh continue to have low wages overall with persistent but variable gender gaps (Jose 2020). Overall, these regional trends suggest that higher wages do not necessarily ensure gender equality, and while some states achieve near parity, significant disparities continue across much of the country. Some states have lower wage discrepancies than others because of variations in economic development, labor market structure, educational attainment, and social policies while states that have more women in the workforce, more diverse economies, and a more robust union presence tend to have smaller total wage gaps due to decreased gender and sectoral wage discrepancies

7. International Policy Frameworks to Reduce the Gender Pay Gap in Agriculture

The following section outlines the key international policy frameworks designed to address and narrow gender pay disparities in the agricultural sector.

Table No. 1 International Policy Frameworks to Reduce the Gender Pay Gap in Agriculture

Policy Framework / Initiative	Year Adopted	Key Provisions	Target Areas	Implementation Mechanism	Impact / Outcomes	Reference
ILO Equal Remuneration	1951	Equal pay for work of equal value, prohibits	Global labour markets	Ratification by member states, national	Ratified by 173 countries, foundation	ILO, 1951

Convention (No. 100)		wage discrimination based on sex	including agriculture	legislation, labour inspections	al for national equal pay laws	
CEDAW	1979	Article 11: Equal pay, Article 14: Elimination of discrimination against rural women	All economic sectors, rural women	State party reporting, CEDAW Committee monitoring, national action plans	Ratified by 189 countries, framework for gender equality legislation	UN Women, 1979
Beijing Platform for Action	1995	Strategic Objective F.1: Promote women's economic rights and independence	Women's economic empowerment including agriculture	National action plans, periodic Beijing+5, +10, +15, +20, +25 reviews	Global blueprint, influenced national policies in 189 countries	UN Women, 1995
Sustainable Development Goals (SDG 5)	2015	Goal 5.4: Recognize unpaid care work, Goal 5.a: Equal rights to resources and land	Gender equality across all sectors	National SDG plans, voluntary reviews, UN monitoring	Universal framework with 169 targets, tracked via 232 indicators	United Nations, 2015
FAO Policy on Gender Equality	2020– 2030	Equal access to resources, services,	Global agrifood systems	Country-level FAO offices,	Framework guiding FAO in	FAO, 2020

		decision-making, address productivity gaps		technical assistance	130+ countries, aims to close 24% productivity gap	
IFAD Gender Policy	2012 (Updated 2022)	Mainstream gender in rural development, access to land, credit, markets	Rural agricultural development	Project design, gender action learning systems, monitoring frameworks	Reached 130M rural people, 47% beneficiaries' women	IFAD, 2012, 2022
UN Women Empowerment Principles	2010	Principle 2: Equal opportunity, Principle 5: Empower women in supply chains	Private sector engagement	Corporate signatories (7,000+), CEO commitment, reporting	Voluntary corporate framework, influence on supply chains	UN Women & UN Global Compact, 2010
ILO Violence and Harassment Convention (No. 190)	2019	First treaty on workplace violence and harassment including gender-based	All workplaces including agriculture	Ratification, national legislation, workplace policies	Ratified by 40 countries (2024), safe workplace standards	ILO, 2019
African Union	2014	Commitment : 10% of national	African agricultural	Biennial review, national	49% of countries on track,	African Union, 2014

Malabo Declaration		budgets to agriculture with gender focus	transformation	agricultural investment plans	gender targets lagging	
Consensus of Brasilia	2010	Autonomy in women's decision-making, land rights for rural women	Rural women in LAC	Regional monitoring via ECLAC, national implementation	Framework for LAC gender policies, influenced 38 countries	ECLAC, 2010
EU Gender Equality Strategy 2020–2025	2020	Pay transparency, binding pay gap measures, intersectionality	All EU sectors including CAP	EU directives, national transposition, ECJ enforcement	Pay gap declined from 16.4% (2010) to 12.7% (2022), agriculture excluded	European Commission, 2020
CAP Gender Mainstreaming	2023–2027	Gender mainstreaming in rural development, women's access to CAP payments	EU agricultural sector	Member state strategic plans, cross-compliance	€387B budget, 29% holdings managed by women, limited gendered data	European Commission, 2023

World Bank Group Gender Strategy	2024–2030	Close participation gaps, improve human capital, address GBV	Global development including agriculture	Country diagnostics, lending conditionalities, results measurement	\$63B committed to gender-tagged operations (FY23)	World Bank, 2024
G20 Women's Financial Inclusion Action Plan	2019	Increase women's access to financial services including agricultural credit	Financial inclusion for women entrepreneurs	National action plans, monitoring dashboard	Reached 1.2B unbanked women, agricultural focus in India & Indonesia	G20, 2019
ILO Decent Work Agenda	1999 (Ongoing)	Rights at work, employment, social protection, dialogue, gender equality crosscutting	All employment sectors globally	National employment policies, ILO cooperation, tripartite dialogue	Adopted by 187 member states, influences global labour standards	ILO, 1999

8. Indian Policy Frameworks to Reduce Gender Pay Gap

The following section outlines the Indian policy frameworks designed to address and narrow gender pay disparities in the agricultural sector.

Table No. 2 Indian Policy Frameworks to Reduce Gender Pay Gap

Policy / Legislation	Year Enacted / Implemented	Key Provisions	Target Beneficiaries	Implementation Mechanism	Budget Allocation / Coverage	Impact / Outcomes	Reference
Minimum Wages Act	1948 (amended regularly)	Fixes minimum wages for agricultural/non-agricultural operations, gender-neutral principle	Agricultural & non-agricultural workers	Ministry of Labour & Employment, state labour departments	N/A (Regulatory)	State variation (₹170–764/day), enforcement gaps allow continued gender discrimination	Govt. of India, 1948, MoL E, 2024
Equal Remuneration Act	1976	Prohibits wage/recruitment discrimination, mandates equal pay	All workers incl. agricultural labourers	Ministry of Labour, inspections, complaints	N/A (Regulatory)	Legal framework for equal pay, weak enforcement in informal agriculture	Govt. of India, 1976
National Commission for Women (NCW)	1992	Investigates violations of women's	All women	MoWCD, suo moto powers	₹30 crore (2023–24)	30,000+ complaints annually, limited agricultural	NCW, 2023

		rights, policy recommen dations				l wage focus	
Kisan Credit Card (KCC) Scheme	1998 (gender focus strengthe ned 2018)	Short- term crop loans, subsidized interest, campaigns for women	Farmers incl. women	MoA, NABARD , banks	₹9.5 lakh crore outstand ing (2023), 7.5 crore cards	Only 13– 15% women cardholder s, land title barrier persists	MoA, 2023
ILO Decent Work Agenda (India adoption)	1999 (ongoing global framewo rk)	Promotes rights at work, employe nt, social protection , dialogue	All employ ment sectors	National employe nt policies, ILO cooperatio n	N/A	Adopted by 187 member states, influences Indian labour standards	ILO, 1999
Mahatma Gandhi NREGA (MGNREGA)	2005	Equal wages, 100 days guarantee d employe nt	Rural househo lds incl. agricult ural workers	MoRD, state implement ation, social audit	₹86,000 crore (FY 2024– 25), 56% women particip ation	Women’s participati on increased from 48% to 56%, wage parity by law but	MoRD, 2024

						gaps persist	
Hindu Succession (Amendment) Act	2005	Equal coparcenary rights for daughters	Hindu women	Ministry of Law, civil courts	N/A (Legislative)	Strengthened property rights, rural enforcement challenges	Govt. of India, 2005
National Food Security Act	2013	Women recognized as household heads for ration cards	Priority/general households (67% population)	MoCA, state food departments	₹2.1 lakh crore (2023–24 subsidy)	81 crore beneficiaries, women's empowerment in household resource control	MoCA, 2013
National Rural Livelihoods Mission (NRLM/DAY)	2011	Mobilizes rural poor women into SHGs, livelihood support	Rural women BPL	MoRD, state missions, SHGs	₹13,335 crore (2023–24), 9 crore women	Women's participation increased, household income increased ₹13k–15k annually	MoRD, 2024
Mahila Kisan Sashaktikaran	2011	Women farmers' rights,	Women farmers	MoRD, state rural	₹1,500 crore (2011–	Productivity increased 20–30%,	MoRD, 2021

Pariyojana (MKSP)		practices, nutrition, capacity building		livelihood missions	21), 1.5M women	income increased 50–100%	
Pradhan Mantri Mudra Yojana (PMMY)	2015	Microfinance loans up to ₹10 lakh, priority to women	Micro-entrepreneurs incl. agri-business women	MoF, banks & MFIs	₹4.57 lakh crore (FY 2023–24), 40.27 crore loans	68% loans to women, avg loan size lower for women	MoF, 2024
Beti Bachao Beti Padhao	2015	Addresses gender discrimination, promotes education & opportunities	Girls & women	MoWCD, multi-sectoral	₹628 crore (2023–24)	Sex ratio improved in 146 districts, indirect impact on agri workforce	MoWCD, 2024
Pradhan Mantri Fasal Bima Yojana (PMFBY)	2016	Crop insurance, women eligible, premium subsidies	Farmers incl. women	MoA, insurance firms	₹15,500 crore (2023–24), 3.6 crore farmers	30–35% women beneficiaries, gaps for tenants/sharecroppers	MoA, 2023

National Policy for Women	2016	Equal access to land, property, resources, empowerment	All women incl. agricultural workers	Women's commissions, ministry coordination	N/A (Policy Framework)	Guiding framework, limited agricultural implementation	MoWCD, 2016
National Agricultural Market (e-NAM)	2016	Electronic trading platform, gender-neutral	All farmers incl. women	MoA, SFAC	1,361 mandis, ₹3.47 lakh crore trade		MoA, 2024
Stand Up India Scheme	2016	Loans ₹10 lakh–₹1 crore for SC/ST & women	Women entrepreneurs incl. agri	MoF, banks	₹43,000 crore, 1.61 lakh accounts (75% women)	Promotes entrepreneurship, limited agri penetration	MoF, 2023
Mission for Integrated Development of Horticulture (MIDH)	2014	Horticulture development, 30% assistance for women	Farmers incl. women horticulturists	MoA, state horticulture depts.	₹2,250 crore (2023–24)	Enhanced subsidy, 8 lakh farmers (25% women)	MoA, 2024
Deen Dayal Upadhyaya Grameen Kaushalya	2014	Rural skill development & placement	Rural youth (15–35) incl. women	MoRD, project agencies	₹4,000 crore (cumulative),	60% women trainees, 70% placement,	MoRD, 2023

Yojana (DDU-GKY)					1.2M trained	limited agri focus	
Periodic Labour Force Survey (PLFS)	2017 (annual)	Gender- disaggreg ated labour force data	Statistic al framew ork	MOSPI	N/A	Evidence base for policies, wage gap documenta tion	MOSPI, Annual
State-Specific Agricultural Wage Notifications	Ongoing	Sets minimum daily wages by gender/tas k	Agricult ural labourer s	State labour depts., district collectors	Varies by state	State- dependent enforceme nt, persistent gaps	State Govts.

9. Conclusion

This chapter provided a holistic understanding of the gender pay gap in agriculture by tracing its historical roots, conceptual foundations, and contemporary manifestations across both developed and developing economies. By examining the evolution of women's roles in farming from subsistence contributors to active economic agents the chapter establishes that their labor has long been undervalued and underrecognized, a pattern deeply embedded in social, cultural, and structural realities. The comparative analysis between developed and developing countries reveals that while the gap is universal, its depth and drivers vary significantly. In developing nations like India, it is compounded by caste hierarchies, land ownership disparities, limited education access, and informal employment norms, whereas in developed economies, occupational segregation and work-life balance penalties remain dominant forces. Despite their crucial role in agricultural production and the growing feminization of the industry, women are still disproportionately employed in low-wage, unpaid, and informal jobs with little access to resources, skills and decision-making authority. Instead of disparities in work effort alone, institutional gender inequality and discrimination are the primary cause of the wage disparity in the agri-food industry. Women are paid less because they are more likely to work in

low-paying, informal agricultural jobs and have limited access to productive resources, education, technology, and decision-making authority.

The discussion of key determinants including human capital differences, bargaining power, institutional discrimination, and policy gaps offers a multidimensional lens through which wage disparities can be understood and addressed. Finally, the review of international frameworks such as ILO conventions and national policies including India's Equal Remuneration Act and MGNREGS underscores that legal and institutional mechanisms, while necessary, remain insufficient without robust implementation, monitoring, and community-level behavioral change. The chapter concludes that the gender pay gap in agriculture is not merely an equity imperative but an economic necessity, one that demands coordinated action across research, policy, and practice.

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